Juniper Networks® vMX Release Notes

Release 18.3R1
21 July 2020

These release notes accompany this release of the Juniper Networks® virtual MX Series router (vMX). They describe new and changed features, limitations, and known problems in the software.

Contents

Introduction

The virtual MX Series router (vMX) is an MX Series router optimized to run on x86 servers. We support Red Hat Enterprise Linux, CentOS, or Ubuntu.

> **NOTE:** See "System Requirements" on page 6 for details on each supported environment.

vMX allows you to leverage Junos OS Release 18.3 to provide quick and flexible deployment. vMX provides the following benefits:

- Optimizes carrier-grade routing for the x86 environment
- Simplifies operations by consistency with MX Series routers
- Introduces new services without reconfiguration of current infrastructure

New and Changed Features

This section describes the new features and enhancements in this release.

- **Class of Service support for configuring maximum number of level 3 scheduler nodes**—Starting in Junos OS Release 18.3R1, to address issues with vMX QOS performance for an interface that has a low number of learning interfaces (IFLs), a new CLI option **maximum-l3-nodes** has been added to **hierarchical-scheduler** to allow you to configure the maximum level 3 nodes for a port.

  The default value for **maximum-l3-nodes** is 16K (16 * 1024). You can configure **maximum-l3-nodes** to a minimum of 2 and a maximum of 32K (32 * 1024). The configured value must be a power of 2.

  For example:

  ```
  user@vmx# set interfaces ge-0/0/0 hierarchical-scheduler maximum-l3-nodes 32768
  ```

  > **NOTE:** This configuration must be present before you reboot the FPC.

  In addition, the scaling number has changed for the **maximum-l2-nodes** CLI option. The maximum level 2 scheduler nodes per port has been changed from 16 to 16K (16 * 1024).

[See Understanding Hierarchical CoS for Subscriber Interfaces.]

- **Nutanix Support for vMX**—Starting in Junos OS Release 18.3R1, you can deploy a vMX instance on a Nutanix Acropolis Hypervisor (AHV) 5.5 cluster. Nutanix AHV is a virtualization solution included with Acropolis that delivers enterprise virtualization ready for multi-cloud environments. Nutanix nodes with AHV include a distributed VM management service responsible for storing VM configuration, making scheduling decisions, and exposing a management interface.
- **EVPN P2MP bud router support (MX Series and vMX)**—Starting in Junos OS Release 18.3R1, Junos OS supports configuring a point-to-multipoint (P2MP) label-switched path (LSP) as a provider tunnel on a bud router. The bud router functions both as an egress router and a transit router.

  To enable a bud router to support P2MP LSP, include the **evpn p2mp-bud-support** statement at the **[edit routing-instances routing-instance-name protocols evpn]** hierarchy level.

  [See Configuring Bud Node Support.]

Known Behavior

This section contains the known behaviors and limitations in this release.

- Scale limitation is observed with VLAN tag operation and circuit cross-connect (CCC).
- When vMX is deployed, the management port is not configured, so you must use the serial console for configuration. Only a small number of configuration lines can be pasted in the vMX console. As a workaround, perform initial configuration to set the root password and to allow SSH access in the vMX console and perform further configuration using SSH.
- ICMP echo request packets are handled inline, which means ICMP packets do not reach the VCP and are replied to by the VFP itself. No ICMP packets are seen on the VCP and packet capture tools do not capture ICMP packets on the VCP.

Known Issues

This section lists the known issues in this release.

- When the FPC is restarted, **kernel: GENCFG: op 32 (Resync blob) failed** syslog message appears. This harmless message can be ignored. PR1050467
- LACP packets are getting dropped on the bridge (for VirtIO). PR1059231
- LLDP packets are getting dropped on the bridge (for VirtIO). PR1066850

- When the FPC is restarting, **COS(cos_chassis_scheduler_pre_add_action:2137)** error messages appear indicating that flexible queuing mode is not enabled. PR1070655
- When the FPC is restarting, a **Received unsupported pic_mask 0x1 ignored message** message appears in the syslog file. PR1072436
- Traffic loss occurs at a remote receiver because of lost remote PIM joins to the local receiver. PR1087031
- When committing the configuration, **%DAEMON-4-SNMP_EVLIB_FAILURE: PFED ran out of transfer credits with PFE.Failed to get stats.** messages sometimes appear in the syslog file. PR1124902
- When committing the configuration, **pfed: %USER-3: downward spike received from pfe for opackets_reply:3545 opackets_record:48932521** messages sometimes appear in the syslog file. PR1146002
- Many **clksync_update_tod_to_pfe:109 Failed to send non-PEC pfe TOD update to other PFEs. Error code: 29.** messages appear in the syslog file. PR1148344
- Scheduler slips are occurring in large VPLS configurations and displaying **JTASK_SCHED_SLIP** log messages. PR1153290
- Observing **rpd[7116]: %DAEMON-6: ifl_delete: ifl** error messages that indicate a multicast tunnel (mt) has been deleted. PR1156725
- CPU pinning is not working properly with OVS bridges. PR1171096
- Intel X710 NICs with 4 x 10G ports might see a slightly lower PPS when all ports are running a 40G line rate. A 40G line rate can be achieved at a 512 byte packet size for the Intel X710 NIC. PR1281366

Subscriber Management

- When bringing up an LNS session on vMX, the Cos-Shaping-Rate attribute (ERX-Attr-177) is sometimes omitted from the Acct-Start messages sent to RADIUS. PR1167154

Resolved Issues

This section lists the issues fixed in this release.

- The **/etc/passwd** file is created in the process of the first commit when a pristine jinstall image is used to boot for the first time. If you configure **event-options**, the system will try to read the configuration from the available event scripts, which requires privileges obtained from the **/etc/passwd** file. This can cause a circular dependency because the commit will not pass if the configuration includes **event-options** the first time a pristine image boots up, which is the case of an upgrade performed with the **virsh create** command. PR1220671
- In a vMX VM with WindRiver Linux 6 RCPL 13, defect LIN6-10388 might be encountered that can result in the disruption of traffic. The issuis has been fixed in the later RCPL versions. PR1351915
- The default NIC adapter type changed from E1000 to VMXNET3. As a result, the default setting on the OVA file for vFPC was not set to the correct NIC driver [VMXNET3, and the vFPC was loaded with the E1000 driver. PR1365337
- While deploying a vMX instance without the virtual control plane (VCP) VM, the script will attempt to allocate vCPUs for the VCP. When there is not enough cores on the host, the script will fail. PR1365921
- In MX150, an upgrade might fail as a result of the the installation of the rpm **nfx-2-routing-data-plane-1.0-0.x86_64** failing. This failure occurs because there is not enough space in **/filesystem** on the host to install the rpm. PR1366324
- Due to an issue with the multicast function on 10-Gigabit XE interfaces, Virtual Router Redundancy Protocol (VRRP) might not work properly If the network configuration includes two vMX VMs running VRRP on a LAN. In this case, both vMX VMs will appear as the master (active). PR1371838
- The vMX might experience issues with QOS performance in an interface that has a low number of learning interfaces (IFLs). To address this vMX QOS performance issue, a new CLI option **maximum-l3-nodes** has been added to **hierarchical-scheduler** to allow you to configure the maximum level 3 nodes for a port. For core interfaces, this number is expected to remain small, and can enhance the scheduler throughput as less scanning cycles are used. See "New and Changed Features" on page 2 for details on class of service support for configuring a maximum number of level 3 scheduler nodes. PR1373999

Upgrade Instructions

You cannot upgrade Junos OS for the vMX router from earlier releases using the **request system software add** command.

You must deploy a new vMX instance using the downloaded software package.

System Requirements

IN THIS SECTION

These topics provide system requirements for each supported environment.

Minimum Hardware and Software Requirements for KVM

Table 1 on page 6 lists the hardware requirements.

Table 1: Minimum Hardware Requirements for KVM

Description	Value
Required CPUs and NICs	For lab simulation and low performance (less than 100 Mbps) use cases, any x86 processor (Intel or AMD) with VT-d capability. For all other use cases, Intel Ivy Bridge processors or later are required. Example of Ivy Bridge processor: Intel Xeon E5-2667 v2 @ 3.30 GHz 25 MB Cache For single root I/O virtualization (SR-IOV) NIC type Intel X520 NICs using ixgbe driver or Intel X710 using i40e driver are required. Only 10G ports are supported. OEM versions of those NICs are also supported when they use the same drivers.

Table 1: Minimum Hardware Requirements for KVM *(continued)*

Description	Value
Number of cores **NOTE:** Performance mode is the default mode and the minimum value is based on one port.	For lite mode with lab simulation use case applications: Minimum of 3 • 1 for VCP • 2 for VFP **NOTE:** If you want to use lite mode when you are running with more than 3 vCPUs for the VFP, you must explicitly configure lite mode.
	For performance mode with low-bandwidth (virtio) or high-bandwidth (SR-IOV) applications: Minimum of 9 • 1 for VCP • 8 for VFP The exact number of vCPUs needed differs depending on the Junos OS features that are configured and other factors, such as average packet size. You can contact Juniper Networks Technical Assistance Center (JTAC) for validation of your configuration and make sure to test the full configuration under load before use in production. For typical configurations, we recommend the following formula to calculate the minimum vCPUs needed by the VFP. **NOTE:** To calculate the optimal number of vCPUs needed by VFP for performance mode: • Without CoS—(4 * *number-of-ports*) + 4 • With CoS—(5 * *number-of-ports*) + 4 **NOTE:** All VFP vCPUs must be in the same physical non-uniform memory access (NUMA) node for optimal performance. In addition to vCPUs for the VFP, we recommend 2 x vCPUs for VCP and 2 x vCPUs for Host OS on any server running the vMX.
Memory **NOTE:** Performance mode is the default mode.	For lite mode: Minimum of 3 GB • 1 GB for VCP • 2 GB for VFP For performance mode: • Minimum of 5 GB 1 GB for VCP 4 GB for VFP • Recommended of 16 GB 4 GB for VCP 12 GB for VFP Additional 2 GB recommended for host OS

Table 1: Minimum Hardware Requirements for KVM *(continued)*

Description	Value
Storage	Local or NAS Each vMX instance requires 30 GB of disk storage
Other requirements	Intel VT-d capability Hyperthreading (recommended) AES-NI

Table 2 on page 8 lists the software requirements for Ubuntu.

Table 2: Software Requirements for Ubuntu

Description	Value
Operating system	Ubuntu 16.04.5 LTS (recommended host OS) Linux 4.4.0-generic
Virtualization	QEMU-KVM 2.0.0+dfsg-2ubuntu1.11
Required packages **NOTE:** Other additional packages might be required to satisfy all dependencies.	bridge-utils qemu-kvm libvirt-bin python python-netifaces vnc4server libyaml-dev python-yaml numactl libparted0-dev libpciaccess-dev libnuma-dev libyajl-dev libxml2-dev libglib2.0-dev libnl-dev python-pip python-dev libxml2-dev libxslt-dev **NOTE:** libvirt 1.3.1

Table 3 on page 8 lists the software requirements for Red Hat Enterprise Linux.

Table 3: Software Requirements for Red Hat Enterprise Linux

Description	Value
Operating system	• Red Hat Enterprise Linux 7.2 Linux 3.10.0-327.4.5 • Starting with Junos OS Release 17.3R1 Red Hat Enterprise Linux 7.3 Linux 3.10.0-514.6.2
Virtualization	QEMU-KVM 1.5.3

Table 3: Software Requirements for Red Hat Enterprise Linux *(continued)*

Description	Value
Required packages **NOTE:** SR-IOV requires these packages: kernel-devel gcc	python27-python-pip python27-python-devel numactl-libs libpciaccess-devel parted-devel yajl-devel libxml2-devel glib2-devel libnl-devel libxslt-devel libyaml-devel numactl-devel redhat-lsb kmod-ixgbe libvirt-daemon-kvm numactl telnet net-tools **NOTE:** libvirt 1.2.17 or later

Table 4 on page 9 lists the software requirements for CentOS.

Table 4: Software Requirements for CentOS

Description	Value
Operating system	CentOS 7.2 Linux 3.10.0-327.22.2
Virtualization	QEMU-KVM 1.5.3
Required packages	python27-python-pip python27-python-devel numactl-libs libpciaccess-devel parted-devel yajl-devel libxml2-devel glib2-devel libnl-devel libxslt-devel libyaml-devel numactl-devel redhat-lsb kmod-ixgbe libvirt-daemon-kvm numactl telnet net-tools **NOTE:** libvirt 1.2.19 To avoid any conflicts, install libvirt 1.2.19 instead of updating from libvirt 1.2.17.

Minimum Hardware and Software Requirements for VMware

Table 5 on page 10 lists the hardware requirements.

Table 5: Minimum Hardware Requirements for VMware

Description	Value
Number of cores **NOTE:** Performance mode is the default mode and the minimum value is based on one port.	For performance mode with low-bandwidth (virtio) or high-bandwidth (SR-IOV) applications: Minimum of 9 • 1 for VCP • 8 for VFP The exact number of vCPUs needed differs depending on the Junos OS features that are configured and other factors, such as average packet size. You can contact Juniper Networks Technical Assistance Center (JTAC) for validation of your configuration and make sure to test the full configuration under load before use in production. For typical configurations, we recommend the following formula to calculate the minimum vCPUs needed by the VFP. **NOTE:** To calculate the optimal number of vCPUs needed by VFP for performance mode: • Without CoS—(4 * *number-of-ports*) + 4 • With CoS—(5 * *number-of-ports*) + 4 **NOTE:** All VFP vCPUs must be in the same physical non-uniform memory access (NUMA) node for optimal performance. In addition to vCPUs for the VFP, we recommend 2 x vCPUs for VCP and 2 x vCPUs for Host OS on any server running the vMX.
	For lite mode: Minimum of 3 • 1 for VCP • 2 for VFP **NOTE:** If you want to use lite mode when you are running with more than 3 vCPUs for the VFP, you must explicitly configure lite mode.
Memory **NOTE:** Performance mode is the default mode.	For performance mode: • Minimum of 5 GB 1 GB for VCP 4 GB for VFP • Recommended of 16 GB 4 GB for VCP 12 GB for VFP For lite mode: Minimum of 3 GB • 1 GB for VCP • 2 GB for VFP

Table 5: Minimum Hardware Requirements for VMware *(continued)*

Description	Value
Storage	Local or NAS Each vMX instance requires 30 GB of disk storage

Table 6 on page 11 lists the software requirements.

Table 6: Software Requirements for VMware

Description	Value
Hypervisor	ESXi 5.5 Update 2
Management Client	vSphere 5.5 or vCenter Server

Requesting Technical Support

Technical product support is available through the Juniper Networks Technical Assistance Center (JTAC). If you are a customer with an active J-Care or Partner Support Service support contract, or are covered under warranty, and need post-sales technical support, you can access our tools and resources online or open a case with JTAC.

- JTAC policies—For a complete understanding of our JTAC procedures and policies, review the *JTAC User Guide* located at https://www.juniper.net/us/en/local/pdf/resource-guides/7100059-en.pdf.
- Product warranties—For product warranty information, visit http://www.juniper.net/support/warranty/.
- JTAC hours of operation—The JTAC centers have resources available 24 hours a day, 7 days a week, 365 days a year.

Self-Help Online Tools and Resources

For quick and easy problem resolution, Juniper Networks has designed an online self-service portal called the Customer Support Center (CSC) that provides you with the following features:

- Find CSC offerings: https://www.juniper.net/customers/support/
- Search for known bugs: https://prsearch.juniper.net/
- Find product documentation: https://www.juniper.net/documentation/
- Find solutions and answer questions using our Knowledge Base: https://kb.juniper.net/
- Download the latest versions of software and review release notes: https://www.juniper.net/customers/csc/software/
- Search technical bulletins for relevant hardware and software notifications: https://kb.juniper.net/InfoCenter/
- Join and participate in the Juniper Networks Community Forum: https://www.juniper.net/company/communities/
- Create a service request online: https://myjuniper.juniper.net

To verify service entitlement by product serial number, use our Serial Number Entitlement (SNE) Tool: https://entitlementsearch.juniper.net/entitlementsearch/

Creating a Service Request with JTAC

You can create a service request with JTAC on the Web or by telephone.

- Visit https://myjuniper.juniper.net.
- Call 1-888-314-JTAC (1-888-314-5822 toll-free in the USA, Canada, and Mexico).

For international or direct-dial options in countries without toll-free numbers, see https://support.juniper.net/support/requesting-support/.

Revision History

21 July 2020—Revision 4, Junos OS Release 18.3R1—vMX

21 February 2019—Revision 3, Junos OS Release 18.3R1—vMX

11 October 2018—Revision 2, Junos OS Release 18.3R1—vMX

26 September 2018—Revision 1, Junos OS Release 18.3R1—vMX